Auteur: Seda Jusjajeva
Studierichting: Informatica
Oefeningenreeks 5A nr. 6.3

$$\int_0^2 \frac{2x}{x^2+1}dx$$
$$= \int \frac{2x}{x^2+1}dx$$
$$= 2\int \frac{x}{x^2+1}dx$$

Substitutie: $t = x^2 + 1 \Rightarrow dt = 2xdx$

$$= \int \frac{1}{t}dt$$
$$= \ln\left|x^2+1\right| + c$$
$$\Rightarrow \int_0^2 \frac{2x}{x^2+1}dx = \left[\ln\left|x^2+1\right|\right]_0^2 = \ln\left|2^2+1\right| - \ln\left|0^2+1\right| = \ln 5 - \ln 1 = \ln 5$$

References